

Meme Stock Scanner

Design, Pipeline, Empirical Findings, and Decisions

Quant research notes

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Abstract

A system to (1) pre-build a watchlist of meme-able stocks from structural characteristics, (2) monitor them daily to detect the moment a name ignites into a meme/short-squeeze move, and (3) enter with intraday timing. This document explains the idea, walks through both notebooks (`meme_squeeze_scanner.ipynb` and `meme_entry_v2.ipynb`), reports the empirical findings from the backtest, and answers the open design and data-access questions. Core conclusion: what is backtestable (daily price/volume) is a trend-follower with a ceiling; the real edge lives in leading, real-time data (borrow/short-interest, social acceleration) that has no free history – so v1 must be lean, gate-by-squeeze, and validated forward.

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1 The Idea and Thesis

The strategy separates two layers with different time constants:

- Structural (slow, scanned once/day): small float, high short interest, small/mid market cap, weak fundamentals. Defines “who can burn” – the candidate watchlist. Market cap is used as a band (floor/ceiling of the universe), never as a trigger: filtering “after mcap already rose” is too late.
- Ignition (fast, looped every few minutes): social acceleration, borrow-fee spikes, RVOL bursts. Defines “who is starting to burn”.

The watchlist is pre-loaded before any price move. When an ignition signal fires on a watchlist name, we are already prepared. An LLM is used only to turn raw social text into structured context (bull/bear regime, narrative, catalysts) – it is a feature, not the decision-maker. The trading decision is rule-based.

2 System Overview

Two notebooks:

- `meme_squeeze_scanner.ipynb` – the full pipeline. Sections 0–7 are the live scanner (setup, universe, borrow, social, LLM, score, trigger, loop). Sections 8–11 are backtest and diagnostics.
- `meme_entry_v2.ipynb` – a simpler, path-dependent absorption entry that reuses the same cached data.

Data / infrastructure

- Have: Alpaca key+secret (paper), OpenRouter → DeepSeek, ApeWisdom (social, keyless), Alpaca RVOL. Reddit / StockTwits are best-effort (flaky).
- Still need: real-time borrow fee / shortable shares. Currently substituted by iBorrowDesk (lagging, shallow history); IBKR API is real-time-only and cannot be backtested.

3 Live Pipeline Walkthrough (Sections 0–7)

S0 – Setup & config. All keys, thresholds, and weights in one `CONFIG` dict. Keys read from environment with fallbacks. A browser-like `User-Agent` is required to avoid 403 bot-blocks on keyless HTTP sources.

S1 – Universe builder (structural, $1\times/\text{day}$). The candidate universe is bounded on purpose (a few thousand names would kill free fundamentals APIs). Three sources:

1. a static seed of historical / plausible meme names across sectors;
2. ApeWisdom trending (names being discussed now);
3. validation that the name is tradable on Alpaca (REST, no `alpaca-py` dependency).

Fundamentals (float, short % of float, days-to-cover, market cap) come from Finviz (primary, more reliable for these fields) with `yfinance` as fallback.

S2 – Borrow & liquidity layer. Borrow fee and shortable shares from iBorrowDesk (keyless substitute for IBKR). RVOL = today’s volume ÷ average lookback. The signal is the change (Δ borrow-fee spike up, shortable down), not the level; a timestamped snapshot store enables the Δ .

S3 – Social ingestion. Mention counts from ApeWisdom (reliable). Raw text for the LLM from Reddit public JSON (phase-2, often 403) and StockTwits (message volume). Two features: `mention_z` (level) and, more importantly, `mention_accel` (how fast mentions are accelerating – acceleration beats level).

S4 – LLM sentiment (context, not decision). DeepSeek (via OpenRouter) summarizes the social text into structured context: market regime (bull/bear), narrative, concepts, catalysts. It is not asked “is this a meme stock” (its judgment there is unreliable); we only read context and let our pipeline decide.

S5 – Composite “meme heat score”. All features are normalized cross-sectionally (z-score), then combined:

$$\begin{aligned} \text{heat} = & w_1 z(\text{short}\%) + w_2 z(\text{dtc}) + w_3 z(\text{util}) + w_4 z(\text{rvol}) + w_5 \text{mention_accel} \\ & + w_6 z(\Delta\text{borrow_fee}) + w_7 (\text{bull} \cdot \text{coord} \cdot \text{fomo}) + w_8 z(\text{otm_call_vol}) \\ & - w_9 \log(\text{float}) - w_{10} \log(\text{mcap}). \end{aligned}$$

Squeeze signals act as a hard gate, not merely additive, to cut false positives (see findings). Names are ranked; the LLM catalyst is shown as the “why”.

S6 – Trigger / alert. Rule-based, no LLM: fire on names whose leg is set (adaptive squeeze/-momentum) and that confirm intraday (RVOL, above VWAP).

S7 – Persistence & loop. Snapshots are persisted point-in-time (for later backtest and for computing Δ). S1 runs each morning; S2–S6 loop every few minutes during the session.

Adaptive thresholds (no magic numbers)

Structural gates are not fixed constants; they are the percentile of today’s cross-section, floored by a winners-feedback loop:

```
def adaptive_threshold(universe_series, q_universe, winner_values=None):
    thr = percentile(universe_series, q_universe) # relative to today
    if winner_values is not None and len(winner_values) >= 3:
        wfloor = percentile(winner_values, winner_floor_pct)
        thr = min(thr, wfloor) # loose enough to have caught past winners
    return thr
```

Principle: risk/mechanical parameters (stops, sizing, data lag) stay FIXED; crowd-relative thresholds (how much short is “high”, how much volume is a “spike”) should be ADAPTIVE, because “high/low” only means something relative to the regime.

4 Backtest and Diagnostics (Sections 8–11)

Setup. Daily bars from Alpaca (IEX, split-adjusted), 2021→now (when the meme phenomenon actually exists). Honest caveat: social and short-interest have no free history, so the backtest only tests the tradable ignition part (RVOL + momentum). Survivorship bias and no transaction costs; treat as illustrative, not live-accurate.

No look-ahead. Every entry decision uses only bars $\leq i$; the exit scans forward as the trade evolves (not as a decision input); short interest is offset by a ~ 12 -day publication lag; meme-event detection is forward-looking but used only for scoring, never to generate trades.

4.1 Filter ablation (Cell 8.2b)

Cumulatively adding filters to the momentum entry:

Config	Trades	Win %	Total P&L	Final
baseline (all off)	661	43.3	\$216,261	\$316,261
+ hard stop -8%	864	32.9	\$323,994	\$423,994
+ follow-through	828	32.9	\$334,670	\$434,670
+ exhaustion cap (.50)	744	33.9	\$419,858	\$519,858

The exhaustion cap (do not buy $> +50\%$ above MA20) is the biggest single lift. Hard stop and follow-through raise P&L even as win-rate falls (cutting losers fast). Tightening the cap below .50 hurts, because meme ignition days often gap $> +30\%$ above MA20 – those are the biggest winners.

4.2 Timing vs. ignition (Cell 8.5)

Timing	n	Win rate	Avg ret	Total P&L
early	26	0.462	+0.060	\$15,718
reactive	332	0.599	+0.186	\$618,307
no-meme	308	0.240	-0.133	$-\$409,657$

The strategy earns on reactive entries (into a confirmed, building run) and bleeds on no-meme false positives (46% of trades). Being “early” via momentum is rare and barely profitable. Median detection lag is +16 days – structurally a trend-follower.

4.3 Event study: what happens before ignition (Cell 11.1)

Averaging all ~ 803 events aligned at ignition ($T = 0$):

T	rvol	dist_60low	atr_pct	vs_ma20
-20	1.576	0.610	0.102	-0.028
-10	2.169	0.826	0.097	-0.011
-7	1.214	0.796	0.095	-0.018
-5	1.206	0.870	0.095	-0.007
-1	7.924	0.896	0.103	+0.053
0	2.942	0.835	0.119	-0.013
+5	1.372	1.136	0.113	+0.063

The path is legible: an early volume thrust ($T \approx -10$, rvol 2.2) $\rightarrow$ a pullback ($T \approx -7..-5$, vs_ma20 dips negative) $\rightarrow$ the real breakout ($T = -1$, rvol 7.9). Volume leads price (Wyckoff accumulation $\rightarrow$ spring $\rightarrow$ markup). Note dist_60low is already +63% at $T = -15$: ignitions fire from names that have already climbed off their lows – “buy near the absolute bottom” is not what the data shows.

4.4 Overfit diagnostics (Section 10)

Because a single parameter can swing total P&L while recent live-like trades (AMC, WEN) lose, Section 10 checks: P&L concentration (does removing the top-5 winners kill the edge?), per-year stability (is it all 2021?), out-of-sample split (does the second half still work?), a robustness surface (plateau vs. sharp spike), and a hidden-factor search. If every daily feature is weak, the edge is not in daily price – it is in the real-time leading layer.

5 Entry v2 – Absorption (meme_entry_v2.ipynb)

A simpler, path-dependent entry that reuses the same cached data. Idea: enter on a day where RVOL spikes super hard but price stays flat (large volume absorbed with no price progress = accumulation), after which dist_60low and vs_ma20 both dip (pullback), while price is curling back up.

```
absorb = any(rvol[k] > 4.0 and abs(ret1[k]) < 0.04 for k in window) # spike + flat
dip = (dist[i] < dist[i-3]) and (vsma[i] < vsma[i-3]) # pullback
early = rvol[i] < 3.0 # not exploded yet
curl = close[i] > close[i-1] # ticking up
if absorb and dip and early and curl and si_gate_ok(sym, date_i):
    enter(close[i]) # then time the fill on minute VWAP reclaim
```

Exit is unchanged: hard stop -8% + ATR trailing (let winners run) + a hold-monitor – the thesis conditions (short fuel, trend) must remain true every day of the hold, not only at entry. Once the daily entry day is identified, minute bars time the actual fill (first reclaim of VWAP). The notebook also studies (i) recall (how many ignitions the signal catches, and which condition is the bottleneck), and (ii) separability of deep-wins ($> 50\%$) vs. stop-outs, including whether they can be separated using only pre-spike data.

6 Key Empirical Findings (summary)

1. Daily momentum entry is a trend-follower, median lag +16 days.
2. Profit comes from reactive trades (+\$618k); no-meme false positives (46% of trades, $-\$409k$) are the main leak. The lever that matters most is precision, not more upside.
3. The exhaustion cap is the biggest single P&L lift; tightening it too far removes explosive winners.
4. The pre-ignition path is volume thrust $\rightarrow$ pullback $\rightarrow$ breakout. Volume leads price. Entering the pullback after the first thrust is the earliest robust daily signal.
5. The genuine early edge (short-covering in progress, social acceleration, options flow) is not in free historical data – so it cannot be backtested, only forward-tested.

7 A. Design-Shaping Decisions

A1. Trading thesis – how does it make money? Answer: (a) long, momentum + squeeze blend – with one refinement: squeeze is a hard gate, not just a score-boost.

- Enter long on RVOL/FOMO ignition (the tradable part).
- Money is in the reactive cohort; “early” has almost no edge.
- The 46% / –\$409k no-meme leak means short-interest/borrow must filter, not merely add points.
- (b) pure-squeeze cannot be validated (no borrow history); (c) short-side is out (halts, hard-to-borrow, all-long platform).

A2. How lean should v1 be? Answer: (a) lean v1 – even leaner: ApeWisdom + Alpaca RVOL + price/VWAP; StockTwits best-effort only.

- Smoke tests: Reddit `.json` = 403, StockTwits = 403/429 (the two most fragile scrapers); ApeWisdom = 200, clean.
- Defer Reddit and borrow-fee to phase 2; ship on what actually runs.

A3. How to validate before real size? Answer: (a) forward paper / shadow.

- Social/borrow have no free history $\Rightarrow$ (b) backfill is infeasible for free.
- Run shadow for several weeks; log scores + would-be trades; and record borrow/social daily $\Rightarrow$ this accrues the very point-in-time history that later enables (b), with no look-ahead.

A4. Where does ingestion + mention-history live? Answer: (a) shared ingestion, per-user execution.

- Scraping is identical across users and rate-limited; duplicating it multiplies rate-limit exposure for zero differentiation.
- One app-level poller maintains watchlist + mention/borrow history + scores in a shared store; isolation is preserved where it matters – execution (per-user Alpaca keys and positions).

8 B. Data-Access Reality Checks

1. Reddit. Public `.json` returned 403 from our IP (even with browser UA and old.reddit fallback). Official API is OAuth, free tier real-time but no history (Pushshift is dead). $\Rightarrow$ defer to phase 2; if used, OAuth not `.json`.
2. StockTwits. 403/429 in testing; no approved key = unstable screen-scrape. $\Rightarrow$ secondary / best-effort, not a dependency.
3. Borrow fee / shortable. IBKR = real-time only, no history; iBorrowDesk = lagging substitute, shallow history. $\Rightarrow$ not a hard requirement for v1; phase-2 live gate; to backtest it, forward-record or pay (Ortex).

4. ApeWisdom. Confirmed – HTTP 200, trending + mentions + mentions_24h_ago, no key. Primary social source.
5. Intraday RVOL. Backtest uses daily bars; live uses Alpaca minute cumulative volume $\div$ time-of-day-adjusted average. Caveat: free Alpaca is IEX (thin for small caps) or SIP delayed 15 minutes; time-of-day adjustment required.

9 Limitations and Validation Plan

- Daily price/volume has a trend-follower ceiling; the alpha-bearing leading signals have no free history.
- v1: lean momentum entry, gated by squeeze structure, exhaustion-capped, hard-stopped, with a hold-monitor.
- Validation: forward paper/shadow for several weeks; a daily snapshot recorder builds point-in-time borrow/social history for a rigorous future backtest.
- Infrastructure: shared ingestion, per-user execution.
- Biggest upgrade if paid: Ortex/Fintel real-time short-interest and utilization – turns the squeeze fuel from “blind” (2-week lag) into “visible as it happens”.